

$$\text{for } \vec{q} \hat{=} G(\vec{x} - \vec{y}) = \int d^3 q G(\vec{q}) e^{i \vec{q} \cdot (\vec{x} - \vec{y})}$$

$$\Rightarrow \frac{1}{q^2 + m^2} \approx \text{const.} \quad \Leftarrow \quad \int d^3 q = \text{const.} \quad \vec{q}^2 \vec{q}$$

cf. Term 11.11

depending on the energy scale & the process of interest,
choose the most general & which respects appropriate symmetries

simplest yet, yet increasingly useful example:

$$S_f = - \vec{\varphi}^T \frac{\nabla^2}{2m} \vec{\varphi} + c (\vec{\varphi}^T \vec{\varphi})^2 + \left\{ \begin{array}{l} \text{all symmetry respecting terms which conserve} \\ \vec{\varphi}, \vec{\nabla}, \vec{r}, \vec{r} \\ \text{rotational} \end{array} \right.$$

$\uparrow$
 $\begin{pmatrix} \varphi_1 \\ \varphi_2 \\ \vdots \\ \varphi_n \end{pmatrix}$ — Gousson fields

technical side:

$$\langle 12 | \varphi_1^* \varphi_2 \varphi_1^* \varphi_2 | 12 \rangle$$

$$\left[\begin{pmatrix} \varphi_1 \\ \varphi_2 \\ \varphi_3 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \varphi_1 \\ \varphi_2 \\ \varphi_3 \end{pmatrix} \right]^2$$

$$(\varphi_1^* \varphi_2 + \varphi_2^* \varphi_1) (\quad)$$

$$\begin{pmatrix} \varphi_1 \\ \varphi_2 \\ \varphi_3 \end{pmatrix} \begin{pmatrix} \varphi_2 + \varphi_3 \\ \varphi_3 \\ 0 \end{pmatrix} = (\varphi_1 \varphi_2 + \varphi_1 \varphi_3 + \varphi_2 \varphi_3)$$

$$\oint \oint \left(\begin{pmatrix} \varphi_1 \\ \varphi_2 \end{pmatrix} \right)^T \left(\begin{pmatrix} \varphi_1 \\ \varphi_2 \end{pmatrix} \right) = \oint \varphi_1^* \varphi_1 + \oint \varphi_2^* \varphi_2$$